CELL BASED SMOOTHED DSG3 (CS-DSG3)

MỘT SỐ KẾT QUẢ PHÂN TÍCH DYNAMIC BÀI TOÁN TẤM DSG

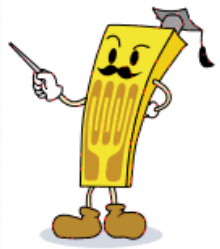
Trình bày: Phùng Văn Phúc

Tp. HCM 30-06-2011

Nội dung trình bày



- Phần tử CS-DSG3
  - Phần tử tấm DSG3
  - Kỹ thuật cell based smoothed cho tấm DSG3 (CS-DSG3)
- Kết quả số
  - Patch test
  - Tấm vuông gối bốn cạnh - SSSS
  - Tấm vuông ngàm bốn cạnh - CCCC

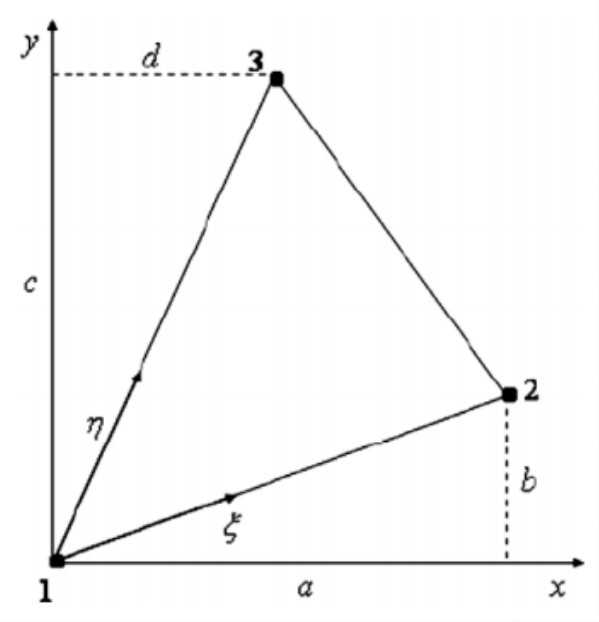


Phần tử CS-DSG3

➢ Phần tử tấm DSG3

○ Bậc tự do mỗi node

$$\mathbf{d}_i^e = \begin{bmatrix} w_i & \beta_{xi} & \beta_{yi} \end{bmatrix}^T \qquad i = 1:3$$



○ Nghiệm xấp xỉ

$$\mathbf{u}^h = \sum_{i=1}^{3} \begin{bmatrix} N_i(\mathbf{x}) & 0 & 0 \\ 0 & N_i(\mathbf{x}) & 0 \\ 0 & 0 & N_i(\mathbf{x}) \end{bmatrix} \mathbf{d}_i = \mathbf{N}\mathbf{d}$$

○ Hàm dạng

$$N_1 = 1 - \xi - \eta \qquad N_2 = \xi \qquad N_3 = \eta$$

Phần tử CS-DSG3

➢ Phần tử tấm DSG3

○ Ma trận độ cứng

$$\mathbf{K}^{DSG3} = \int_{\Omega^e} \left(\mathbf{B}^b\right)^T \mathbf{D}^b \mathbf{B}^b d\Omega + \int_{\Omega^e} \left(\mathbf{B}^s\right)^T \mathbf{D}^s \mathbf{B}^s d\Omega$$

$$= \left(\mathbf{B}^b\right)^T \mathbf{D}^b \mathbf{B}^b A^e + \left(\mathbf{B}^s\right)^T \mathbf{D}^s \mathbf{B}^s A^e$$

$$\mathbf{D}^s = \frac{kt^3}{t^2 + \alpha h_e^2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad k = \frac{5E}{12(1+\nu)}$$

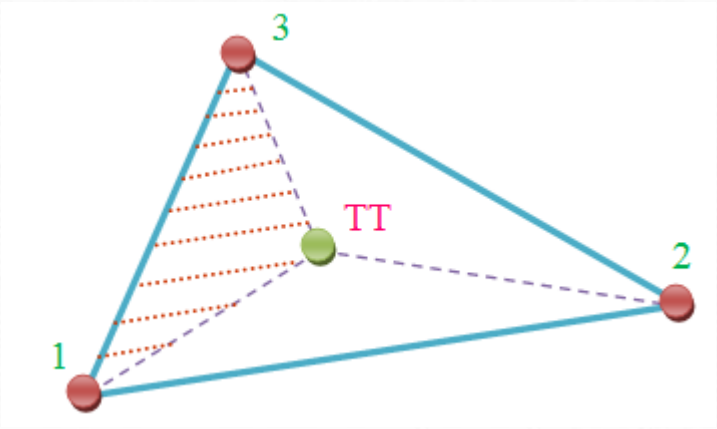
$$\mathbf{D}^b = \frac{Et^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

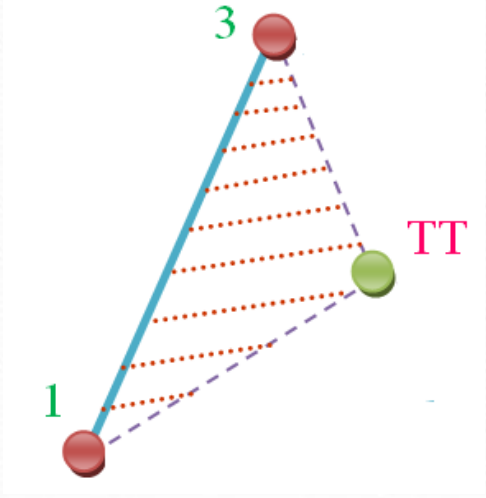
$$\mathbf{B}^b = \frac{1}{2A^e} \begin{bmatrix} 0 & b-c & 0 \\ 0 & 0 & d-a \\ 0 & d-a & b-c \end{bmatrix} \begin{bmatrix} 0 & c & 0 \\ 0 & 0 & -d \\ 0 & -d & c \end{bmatrix} \begin{bmatrix} 0 & -b & 0 \\ 0 & 0 & a \\ 0 & a & -b \end{bmatrix} = \begin{bmatrix} \mathbf{B}_1^b & \mathbf{B}_2^b & \mathbf{B}_3^b \end{bmatrix}$$

$$\mathbf{B}^s = \frac{1}{2A^e} \begin{bmatrix} b-c & A^e & 0 \\ d-a & 0 & A^e \end{bmatrix} \begin{bmatrix} c & \frac{ac}{2} & \frac{bc}{2} \\ -d & -\frac{ad}{2} & -\frac{bd}{2} \end{bmatrix} \begin{bmatrix} -b & -\frac{bd}{2} & -\frac{bc}{2} \\ a & \frac{ad}{2} & \frac{ac}{2} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_1^s & \mathbf{B}_2^s & \mathbf{B}_3^s \end{bmatrix}$$

Phần tử CS-DSG3

➢ Kỹ thuật cell based smoothed DSG3 (CS-DSG3)





○ Phần tử tam giác ban đầu 1-2-3

○ Phần tử tam giác "con" 3-1-TT

1 phần tử T3 → Xử lý phần tử → 3 phần tử T3 "con" (sub-T3)

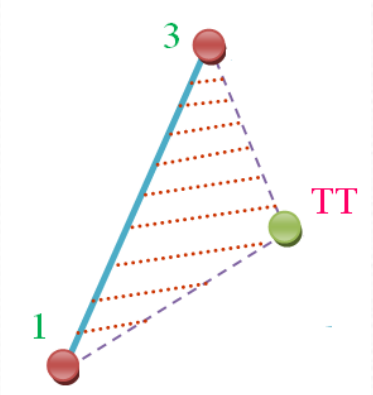
Cải tiến DSG bằng xử lý phần tử

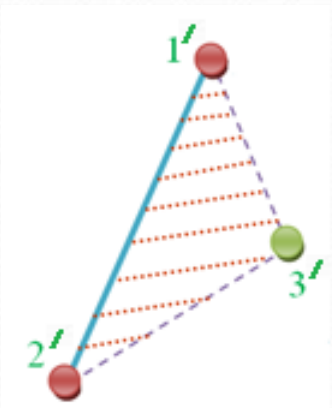
Phần tử CS-DSG3



➢ Kỹ thuật cell based smoothed DSG3 (CS-DSG3)

Global



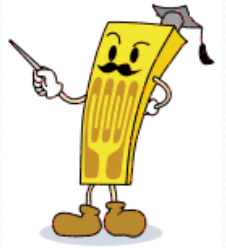


Local

○ Biến dạng trên phần tử sub-T3

$$\boldsymbol{\kappa}^e = \begin{bmatrix} \mathbf{B}_1^b & \mathbf{B}_2^b & \mathbf{B}_3^b \end{bmatrix} \begin{bmatrix} \mathbf{d}_1' \\ \mathbf{d}_2' \\ \mathbf{d}_3' \end{bmatrix} = \mathbf{B}_1^b \mathbf{d}_3 + \mathbf{B}_2^b \mathbf{d}_1 + \mathbf{B}_3^b \mathbf{d}_{TT} = \mathbf{B}_1^b \mathbf{d}_3 + \mathbf{B}_2^b \mathbf{d}_1 + \mathbf{B}_3^b \left(\frac{\mathbf{d}_1 + \mathbf{d}_2 + \mathbf{d}_3}{3} \right)$$

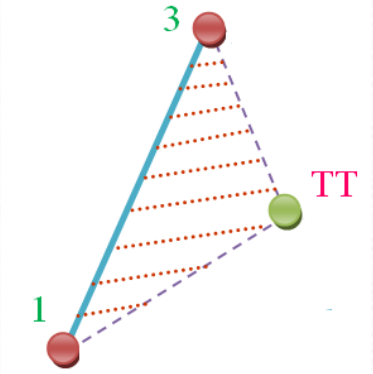
$$= \left(\mathbf{B}_2^b + \frac{\mathbf{B}_3^b}{3} \right) \mathbf{d}_1 + \frac{\mathbf{B}_3^b}{3} \mathbf{d}_2 + \left(\mathbf{B}_1^b + \frac{\mathbf{B}_3^b}{3} \right) \mathbf{d}_3 = \underbrace{\begin{bmatrix} \mathbf{B}_2^b + \frac{\mathbf{B}_3^b}{3} & \frac{\mathbf{B}_3^b}{3} & \mathbf{B}_1^b + \frac{\mathbf{B}_3^b}{3} \end{bmatrix}}_{\mathbf{b}_1^{be}} \begin{bmatrix} \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{bmatrix}$$

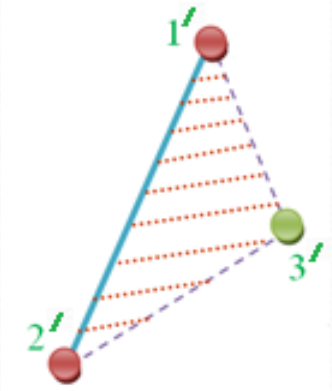


Phần tử CS-DSG3

➢ Kỹ thuật cell based smoothed DSG3 (CS-DSG3)

Global



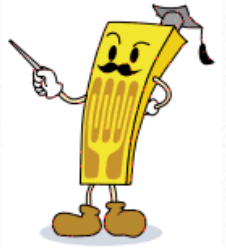


Local

○ Biến dạng trên phần tử sub-T3

$$\boldsymbol{\gamma}^e = \begin{bmatrix} \mathbf{B}_1^s & \mathbf{B}_2^s & \mathbf{B}_3^s \end{bmatrix} \begin{bmatrix} \mathbf{d}_1' \\ \mathbf{d}_2' \\ \mathbf{d}_3' \end{bmatrix} = \mathbf{B}_1^s\mathbf{d}_3 + \mathbf{B}_2^s\mathbf{d}_1 + \mathbf{B}_3^s\mathbf{d}_{TT} = \mathbf{B}_1^s\mathbf{d}_3 + \mathbf{B}_2^s\mathbf{d}_1 + \mathbf{B}_3^s\left(\frac{\mathbf{d}_1 + \mathbf{d}_2 + \mathbf{d}_3}{3}\right)$$

$$= \left(\mathbf{B}_2^s + \frac{\mathbf{B}_3^s}{3}\right)\mathbf{d}_1 + \frac{\mathbf{B}_3^s}{3}\mathbf{d}_2 + \left(\mathbf{B}_1^s + \frac{\mathbf{B}_3^s}{3}\right)\mathbf{d}_3 = \underbrace{\begin{bmatrix} \mathbf{B}_2^s + \frac{\mathbf{B}_3^s}{3} & \frac{\mathbf{B}_3^s}{3} & \mathbf{B}_1^s + \frac{\mathbf{B}_3^s}{3} \end{bmatrix}}_{\mathbf{b}_1^{se}} \begin{bmatrix} \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{bmatrix}$$

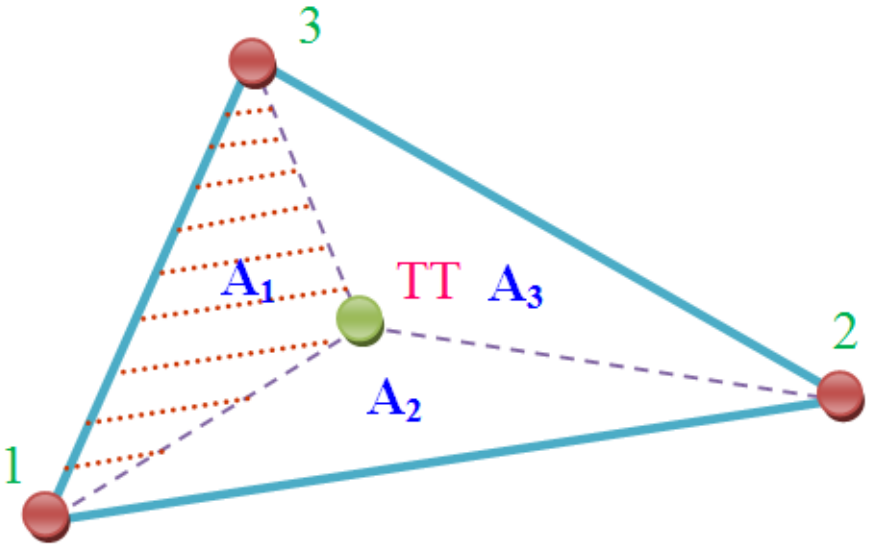


Phần tử CS-DSG3

- Kỹ thuật cell based smoothed DSG3 (CS-DSG3)
  - Áp dụng kỹ thuật "trơn hóa"

$$\tilde{\mathbf{B}}^s = \frac{\mathbf{b}_1^{se}A_1 + \mathbf{b}_2^{se}A_2 + \mathbf{b}_3^{se}A_3}{A_1 + A_2 + A_3} = \frac{\sum_{i=1}^{3}\mathbf{b}_i^{se}A_i}{\sum_{i=1}^{3}A_i}$$

$$\tilde{\mathbf{B}}^b = \frac{\mathbf{b}_1^{be}A_1 + \mathbf{b}_2^{be}A_2 + \mathbf{b}_3^{be}A_3}{A_1 + A_2 + A_3} = \frac{\sum_{i=1}^{3}\mathbf{b}_i^{be}A_i}{\sum_{i=1}^{3}A_i}$$

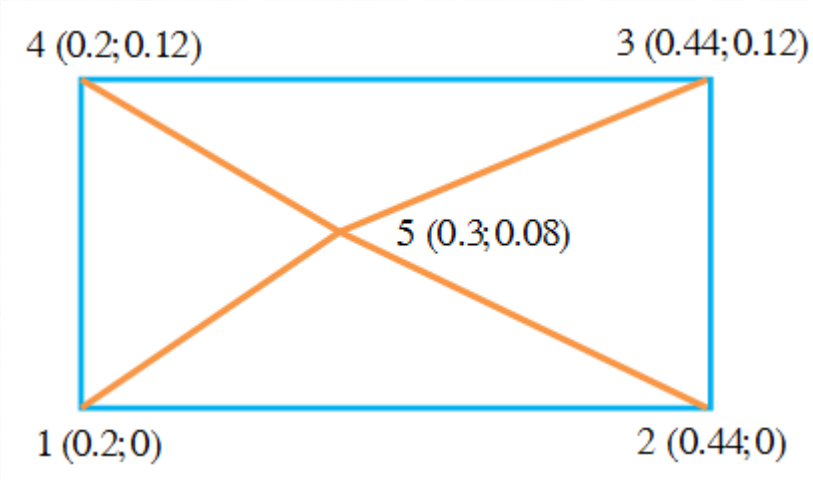


$$\mathbf{K}^{CS-DSG3} = \int_{\Omega^e}\left(\tilde{\mathbf{B}}^b\right)^T\mathbf{D}^b\tilde{\mathbf{B}}^b d\Omega + \int_{\Omega^e}\left(\tilde{\mathbf{B}}^s\right)^T\mathbf{D}^s\tilde{\mathbf{B}}^s d\Omega$$

$$= \left(\tilde{\mathbf{B}}^b\right)^T\mathbf{D}^b\tilde{\mathbf{B}}^b A^e + \left(\tilde{\mathbf{B}}^s\right)^T\mathbf{D}^s\tilde{\mathbf{B}}^s A^e$$

Kết quả số

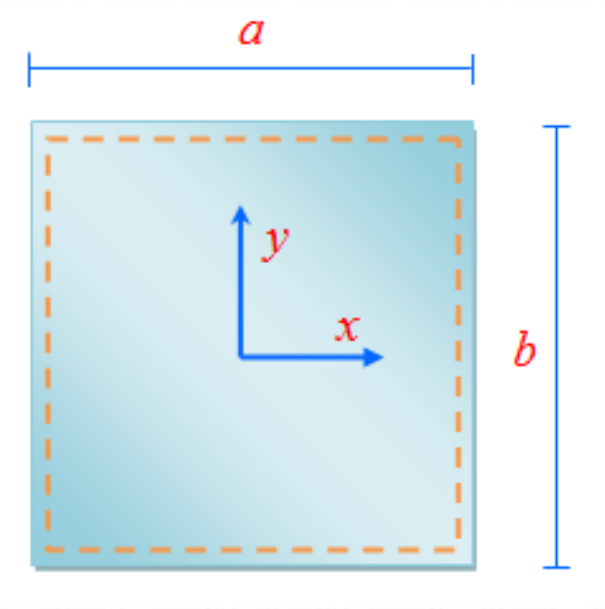
- Patch test



Element	w_5	θ_{x5}	θ_{y5}	m_{x5}	m_{y5}	m_{xy5}
CS-DSG3	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033
Exact	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033

Kết quả số

- Phân tích dao động tấm DSG 4 cạnh gối tựa (SSSS)

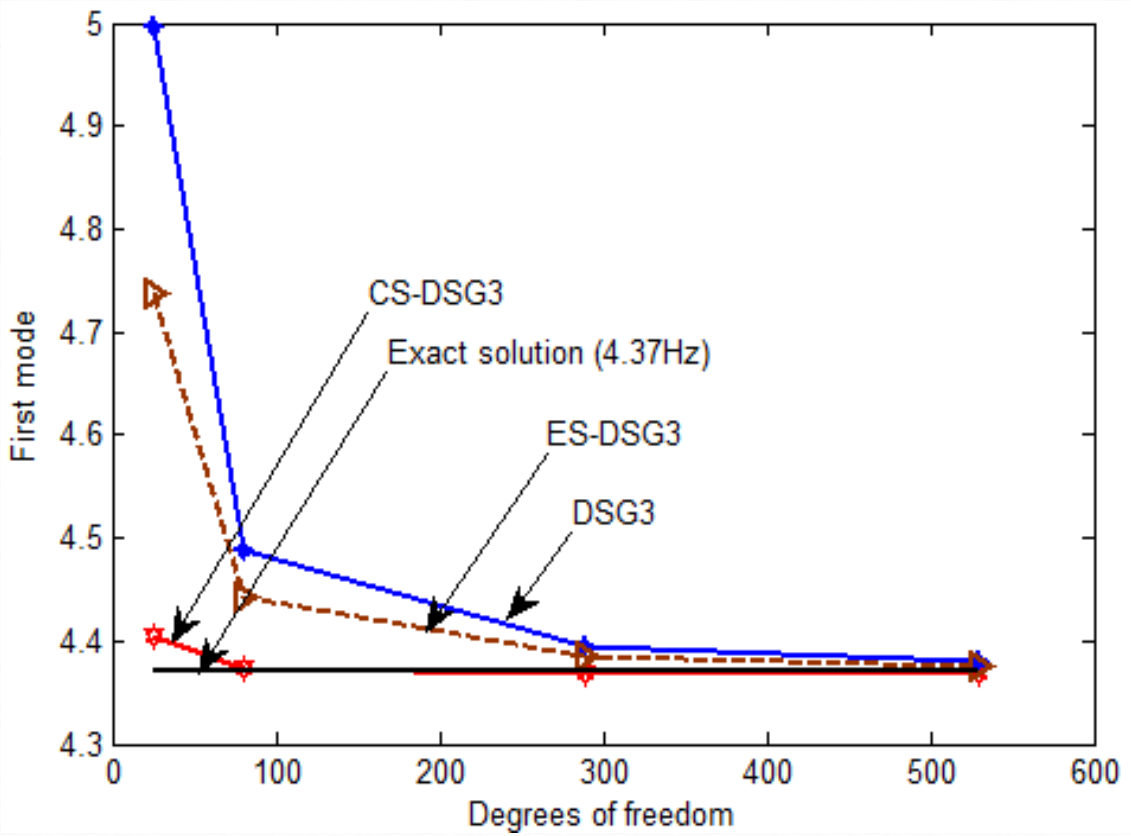


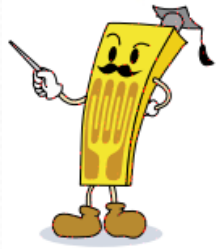
- Trong đó $a/b = 1,\ t/a = 0.1$
- Phân tích dao động tự do $\det(-\omega^2\mathbf{M}+\mathbf{K})=0$

Kết quả số



➢ Phân tích dao động tấm DSG 4 cạnh gối tựa (SSSS)

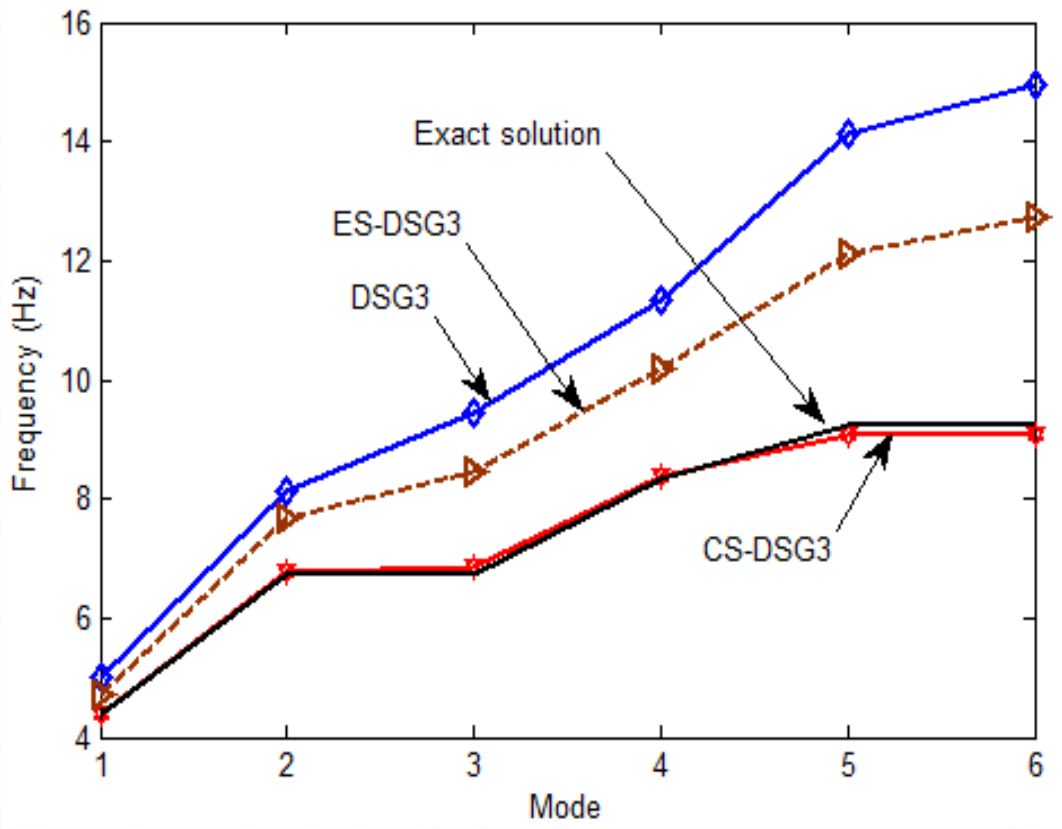




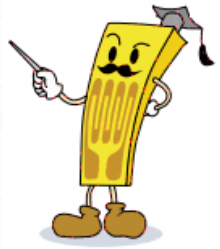
Kết quả số



➢ Phân tích dao động tấm DSG 4 cạnh gối tựa (SSSS)



6 mode đầu với **lưới rất thô** 4 x 4



Kết quả số



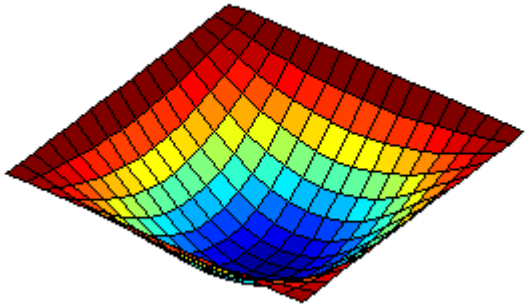
➢ Phân tích dao động tấm DSG 4 cạnh gối tựa (SSSS)

Meshing	Element	Mode sequence number					
		1	2	3	4	5	6
4 x 4	DSG3	4.997	8.149	9.4311	11.354	14.129	14.9353
	ES-DSG3	4.7376	7.658	8.4524	10.1882	12.1227	12.7533
	CS-DSG3	4.4032	6.7790	6.8435	8.3901	9.0714	9.0889
8 x 8	DSG3	4.4891	7.0697	7.253	9.1263	10.2195	10.3361
	ES-DSG3	4.4433	6.9495	7.0727	8.8487	9.8575	9.9221
	CS-DSG3	4.3743	6.7560	6.7712	8.3830	9.2329	9.2341
16 x 16	DSG3	4.3943	6.8227	6.8587	8.5447	9.4557	9.4616
	ES-DSG3	4.3846	6.7922	6.8196	8.4744	9.3666	9.3698
	CS-DSG3	4.3683	6.7470	6.7511	8.3623	9.2256	9.2257
22 x 22	DSG3	4.3809	6.7854	6.8037	8.4543	9.3441	9.3457
	ES-DSG3	4.3759	6.7692	6.7834	8.4173	9.2968	9.2976
	CS-DSG3	4.3674	6.7456	6.7478	8.3587	9.2238	9.2238
	Exact [68]	4.37	6.74	6.74	8.35	9.22	9.22

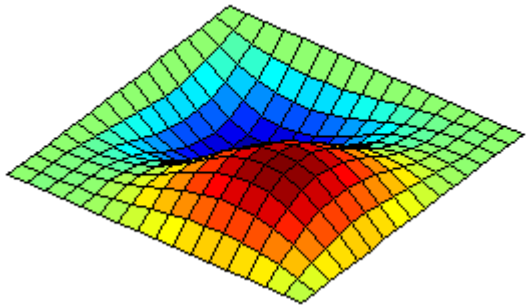
Kết quả số

- Phân tích dao động tấm DSG 4 cạnh gối tựa (SSSS)

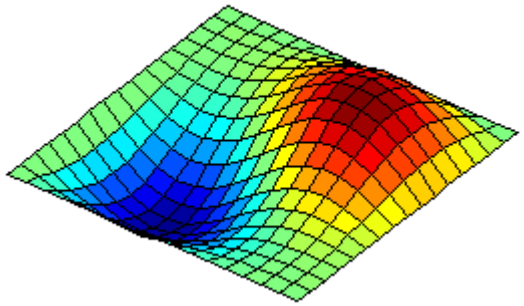
Mode 1 (ω = 4.3683)



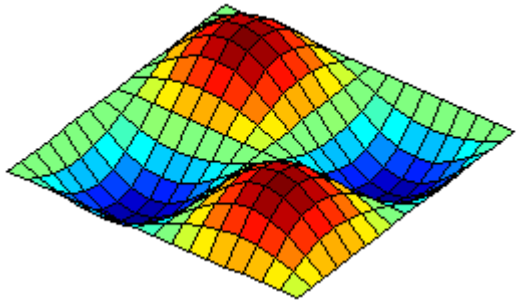
Mode 2 (ω = 6.747)



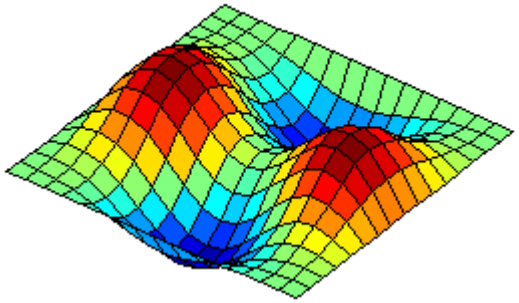
Mode 3 (ω = 6.7511)



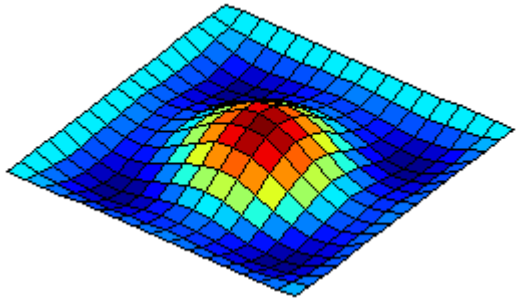
Mode 4 (ω = 8.3623)



Mode 5 (ω = 9.2256)



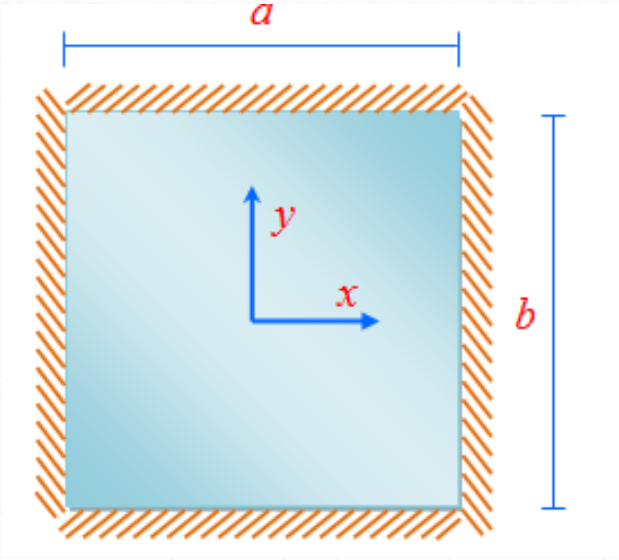
Mode 6 (ω = 9.2257)



Kết quả số

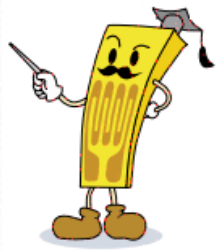


➢ Phân tích dao động tấm DSG 4 cạnh ngàm (CCCC)



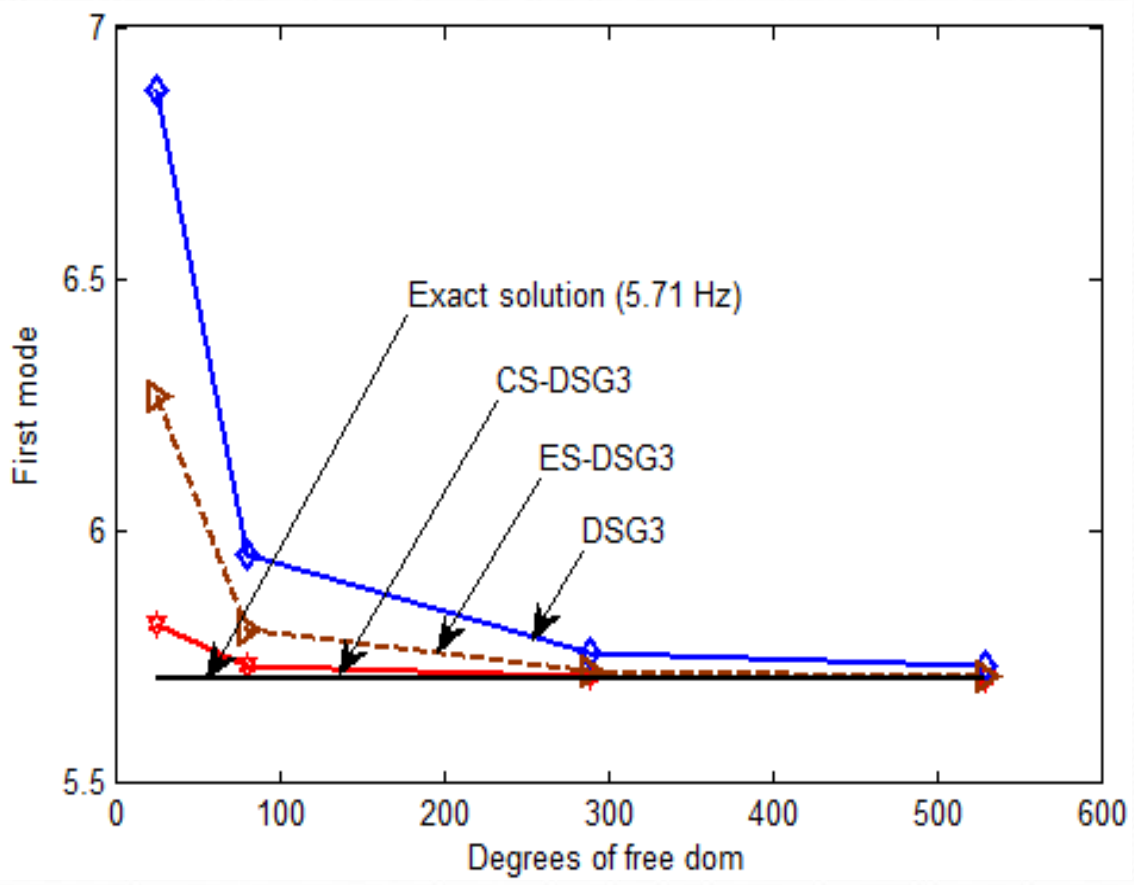
○ Trong đó $a/b = 1,\ t/a = 0.1$

○ Phân tích dao động tự do $\det(-\omega^2\mathbf{M}+\mathbf{K})=0$



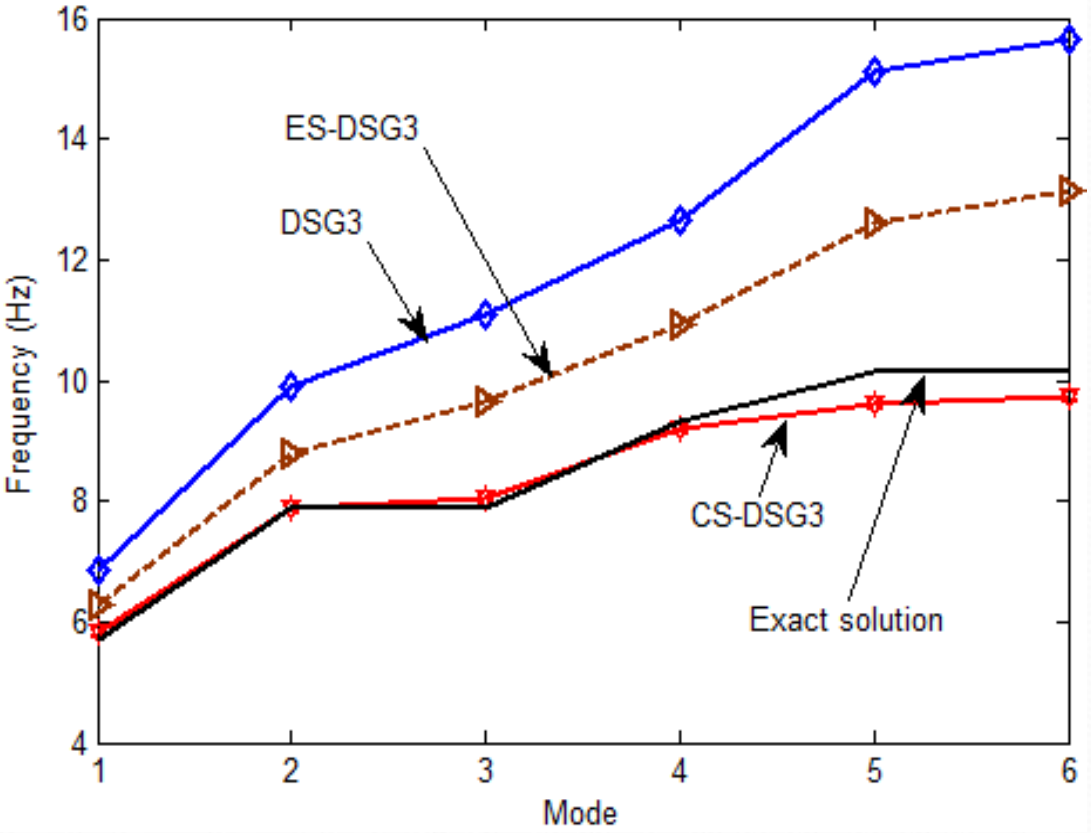
Kết quả số

➢ Phân tích dao động tấm DSG 4 cạnh ngàm (CCCC)



Kết quả số

➢ Phân tích dao động tấm DSG 4 cạnh ngàm (CCCC)



6 mode đầu với **lưới rất thô** 4 x 4

Kết quả số



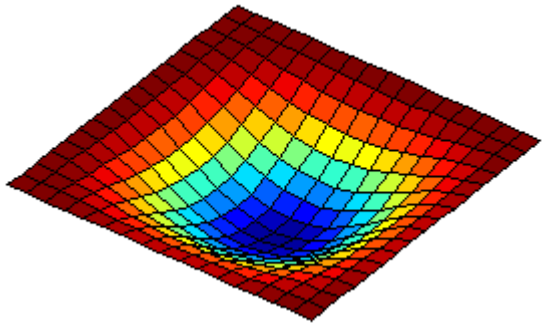
➢ Phân tích dao động tấm DSG 4 cạnh ngàm (CCCC)

Meshing	Element	Mode sequence number					
		1	2	3	4	5	6
4 x 4	DSG3	6.8748	9.8938	11.0847	12.6362	15.1032	15.6402
	ES-DSG3	6.2662	8.7952	9.6625	10.9112	12.6101	13.1360
	CS-DSG3	5.8163	7.8647	8.0481	9.2126	9.6233	9.7156
8 x 8	DSG3	5.9547	8.3618	8.6293	10.2985	11.3415	11.5397
	ES-DSG3	5.8068	8.0861	8.2701	9.8397	10.7600	10.8960
	CS-DSG3	5.7180	7.8901	7.9111	9.3563	10.1380	10.1875
16 x 16	DSG3	5.7616	7.9935	8.0525	9.5772	10.4153	10.4697
	ES-DSG3	5.7250	7.9211	7.9627	9.4499	10.2631	10.3126
	CS-DSG3	5.7117	7.8846	7.8965	9.3442	10.1319	10.1803
22 x 22	DSG3	5.7337	7.9381	7.9686	9.4589	10.2758	10.3246
	ES-DSG3	5.7141	7.8990	7.9206	9.3896	10.1935	10.2411
	CS-DSG3	5.7077	7.8810	7.8873	9.3357	10.1273	10.1754
	Exact [69]	5.7100	7.8800	7.8800	9.3300	10.1300	10.1800

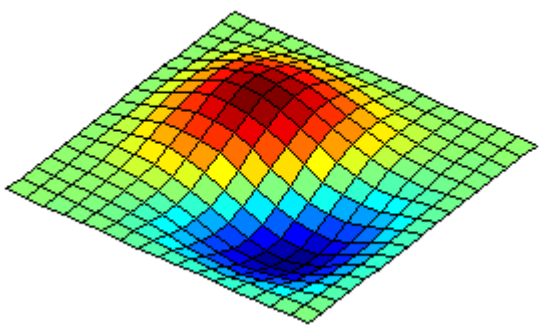
Kết quả số

- Phân tích dao động tấm DSG 4 cạnh ngàm (CCCC)

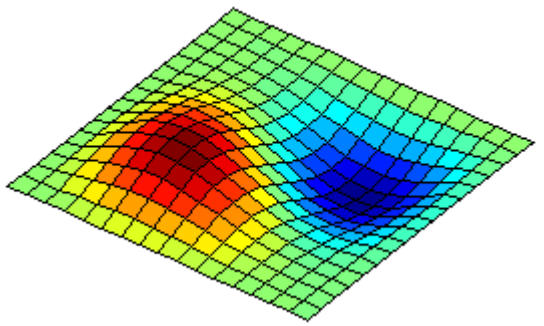
Mode 1 (ω = 5.7117)



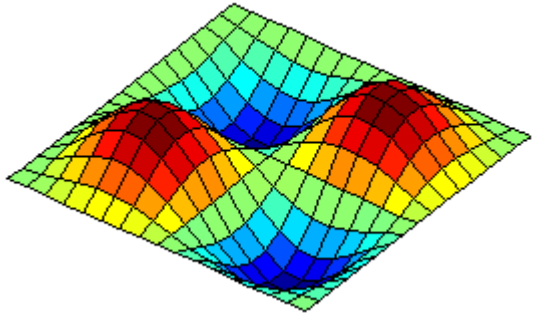
Mode 2 (ω = 7.8846)



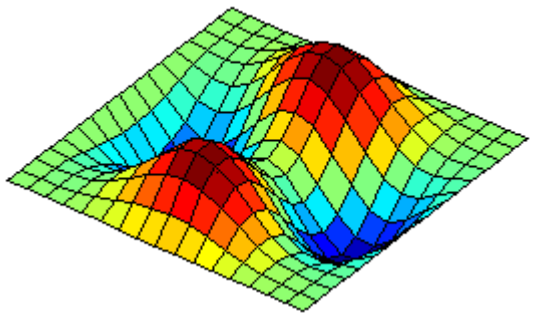
Mode 3 (ω = 7.8965)



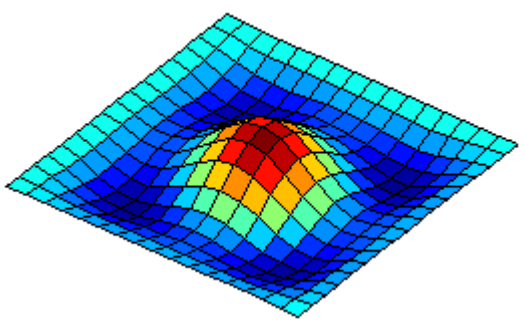
Mode 4 (ω = 9.3442)



Mode 5 (ω = 10.1319)



Mode 6 (ω = 10.1803)



THANKS YOUR ATTENTION

